The Rational Generating Function of A308620

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October 26, 2026

Abstract

We derive the generating function conjectured for the steady-state counts of the evolutionary spatial prisoner's dilemma on a path. The proof starts from the established exact count, rather than a fit to finitely many values. It proves the proposed order-ten recurrence with its correct starting index: the recurrence applies from index 12, and fails by one at index 11.

1 The exact counting formula

Write $a(n)$ for A308620 [1], $n \geq 1$. The counts are for path states modulo reflection. Burger, van der Merwe and van Vuuren prove the following formula in Theorem 3, equation (4.2) of [2]; it is also the entry's Maple evaluator:

$$a(n) = 1 + Q_n + \tfrac{1}{2}(R_n + \bar{R}_n + S_n + \bar{S}_n),$$

where, with empty sums zero,

$$Q_n = \sum_{i=1}^{\lfloor (n+1)/5 \rfloor} \binom{n-3i}{2i-1},$$

$$R_n = \sum_{i=1}^{\lfloor (n+4)/5 \rfloor} \binom{n-3i+2}{2i-2}, \qquad \bar{R}_n = \sum_{i=1}^{\lfloor (n+4)/5 \rfloor} \binom{\lfloor (n-5i+4)/2 \rfloor + i - 1}{i-1},$$

$$S_n = \sum_{i=1}^{\lfloor (n-2)/5 \rfloor} \binom{n-3i-2}{2i}, \qquad \bar{S}_n = \sum_{i=1}^{\lfloor (n-2)/5 \rfloor} \binom{\lfloor (n-5i-2)/2 \rfloor + i}{i}.$$

This established theorem supplies the connection to the game: nontrivial steady states have defection runs of length at least two, internal cooperation runs of length at least three, and endpoint cooperation runs of length at least two. The terms above count their run-length compositions and reflection-fixed cases. We prove the subsequent generating-function conjecture, not a new claim of authorship for this state classification.

2 Summation and recurrence

Put $D = (1-x)^2 - x^5$, $E = 1 - x^2 - x^5$, and let each capital letter also denote the generating function over $n \geq 1$ of its subscripted sequence. The elementary identity $\sum_{j \geq 0} \binom{j+h}{h} x^j = (1-x)^{-h-1}$ gives

$$Q = \sum_{i \geq 1} \frac{x^{5i-1}}{(1-x)^{2i}} = \frac{x^4}{D}, \qquad R = \frac{x(1-x)}{D}, \qquad S = \frac{x^7}{(1-x)D},$$

$$\bar{R} = \frac{x(1+x)}{E}, \qquad \bar{S} = \frac{x^7}{(1-x)E}.$$

For the barred sums, pair the two powers x^{2j} and x^{2j+1} to obtain $\sum_{j \geq 0} \binom{\lfloor j/2 \rfloor + h}{h} x^j = (1+x)/(1-x^2)^{h+1}$. All the sums are formal: every coefficient receives finitely many contributions, so rearrangement requires no analytic convergence.

Theorem 1. *The ordinary generating function is exactly*

$$G(x) = \sum_{n \geq 1} a(n)x^n = 2x - \frac{x^2(-2 + 2x + x^2 - 2x^3 + 3x^5 + x^7 + x^9)}{(x^5 + x^2 - 1)(x^5 - x^2 + 2x - 1)}.$$

For every $n \geq 12$,

$$a(n) = 2a(n-1) - 2a(n-3) + a(n-4) + 2a(n-5) - 2a(n-6) - a(n-10).$$

At $n = 11$ *the right side is 41 whereas* $a(11) = 42$.

Proof. Substitute the five rational expressions above into $G = x/(1-x) + Q + (R + \bar{R} + S + \bar{S})/2$ and combine fractions. An equivalent polynomial identity, which also certifies the algebra, is

$$\begin{aligned}(1 - 2x + 2x^3 - x^4 - 2x^5 + 2x^6 + x^{10})G(x) \\ = 2x - 2x^2 - 2x^3 + 3x^4 - 4x^6 + x^7 - x^9 + x^{11}.\end{aligned}$$

The denominator is DE. Coefficients above degree 11 give the recurrence. The coefficient at degree 11 is one, explaining its explicit exception there. This proves both official conjectures with the required initial-index qualification. □

Validation. The verifier computes the original binomial formula through $n = 1000$, checks every recurrence instance and the entire rational polynomial identity in that range, and compares all 40 published terms. A separate binary-word enumeration through length 18 checks the run conditions and counts reversal orbits directly, independently of the binomial formula. Finite checks validate the implementation; the formal identities prove the all-index claims. No OEIS submission or literature-priority claim is made.

References

[1] R. J. Mathar, OEIS A308620, https://oeis.org/A308620, accessed October 26, 2026.

[2] A. P. Burger, M. van der Merwe and J. H. van Vuuren, An asymptotic analysis of the evolutionary spatial prisoner's dilemma on a path, Discrete Applied Mathematics 160 (2012), 2075–2088, Theorem 3 and equation (4.2), author-hosted copy: https://www.vuuren.co.za/papers/GamesPaths.pdf.